Correlation and Linear Regression

Chapter 13

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Learning Objectives

- LO13-1 Explain the purpose of correlation analysis
- LO13-2 Calculate a correlation coefficient to test and interpret the relationship between two variables
- LO13-3 Apply regression analysis to estimate the linear relationship between two variables
- LO13-4 Evaluate the significance of the slope of the regression equation
- LO13-5 Evaluate a regression equation's ability to predict using the standard estimate of the error and the coefficient of determination
- LO13-6 Calculate and interpret confidence and prediction intervals
- LO13-7 Use a log function to transform a nonlinear relationship

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What is Correlation Analysis?

- Used to report the relationship between two variables

> CORRELATION ANALYSIS A group of techniques to measure the relationship between two variables.

- In addition to graphing techniques, we'll develop numerical measures to describe the relationships

Examples

- Does the amount Healthtex spends per month on training its sales force affect its monthly sales?
- Does the number of hours students study for an exam influence the exam score?

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Scatter Diagram

- A scatter diagram is a graphic tool used to portray the relationship between two variables
- The independent variable is scaled on the X-axis and is the variable used as the predictor
- The dependent variable is scaled on the Y-axis and is the variable being estimated

Sales Representative	Sales Calls	Copiers Sold
Brian Virost	96	41
Carlos Ramirez	40	41
Carol Saia	104	51
Greg Fish	128	60
Jeff Hall	164	61
Mark Reynolds	76	29
Meryl Rumsey	72	39
Mike Kiel	80	50
Ray Snarsky	36	28
Rich Niles	84	43
Ron Broderick	180	70
Sal Spina	132	56
Soni Jones	120	45
Susan Welch	44	31
Tom Keller	84	30

Graphing the data in a scatter diagram will make the relationship between sales calls and copiers sales easier to see.

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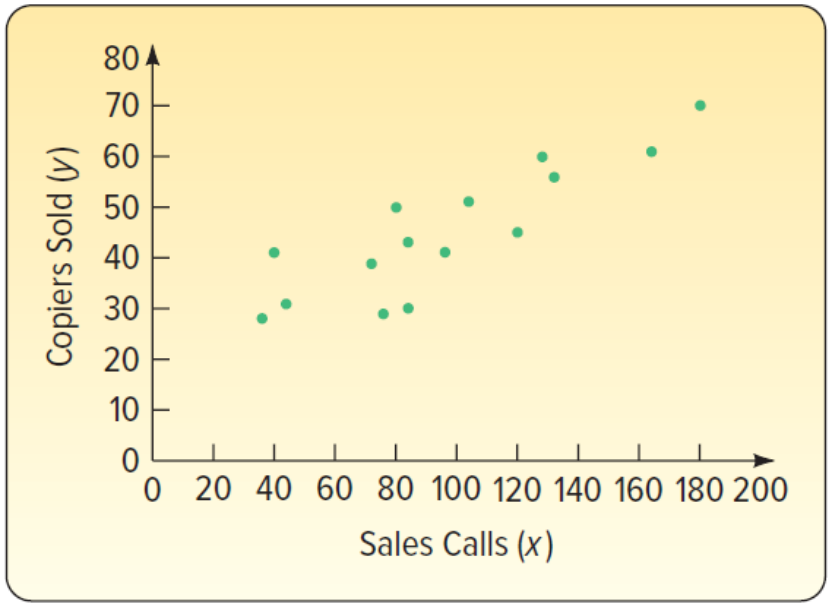
Scatter Diagram Example

North American Copier Sales sells copiers to businesses of all sizes throughout the United States and Canada. The new national sales manager is preparing for an upcoming sales meeting and would like to impress upon the sales representatives the importance of making an extra sales call each day. She takes a random sample of 15 sales representatives and gathers information on the number of sales calls made last month and the number of copiers sold. Develop a scatter diagram of the data.



Sales reps who make more calls tend to sell more copiers!

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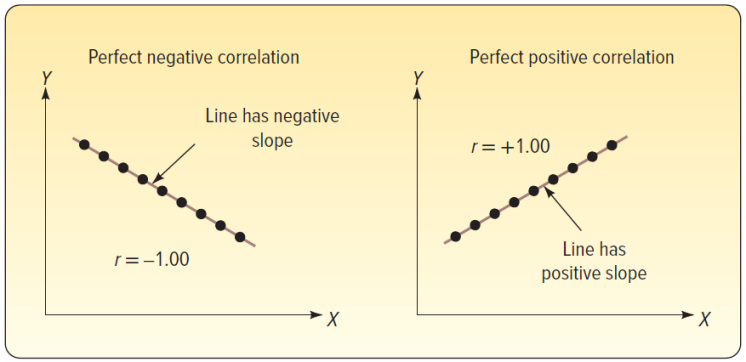
Correlation Coefficient

CORRELATION COEFFICIENT A measure of the strength of the linear relationship between two variables.

- Characteristics of the correlation coefficient are:
  - The sample correlation coefficient is identified as r
  - It shows the direction and strength of the linear relationship between two interval- or ratio-scale variables
  - It ranges from −1.00 to 1.00
  - If it's 0, there is no association
  - A value near 1.00 indicates a direct or positive correlation
  - A value near −1.00 indicates a negative correlation



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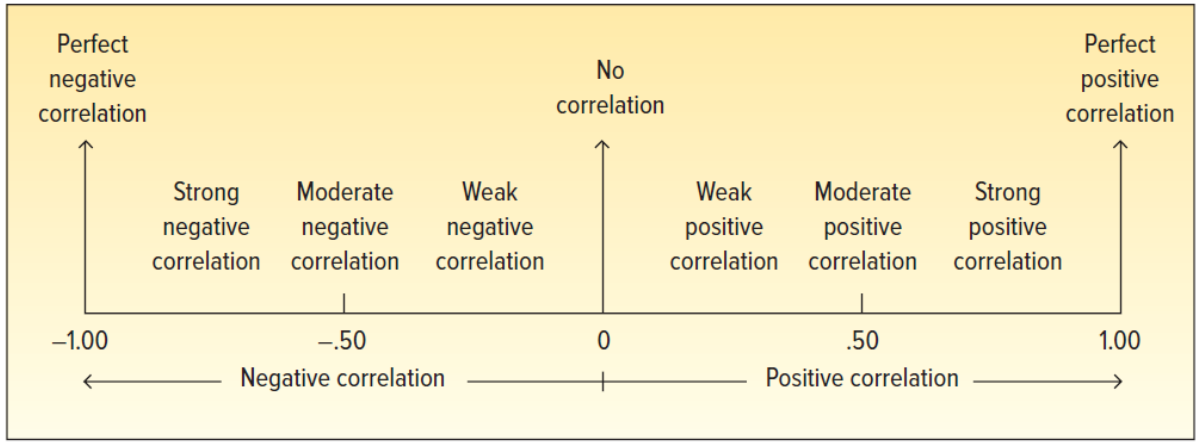
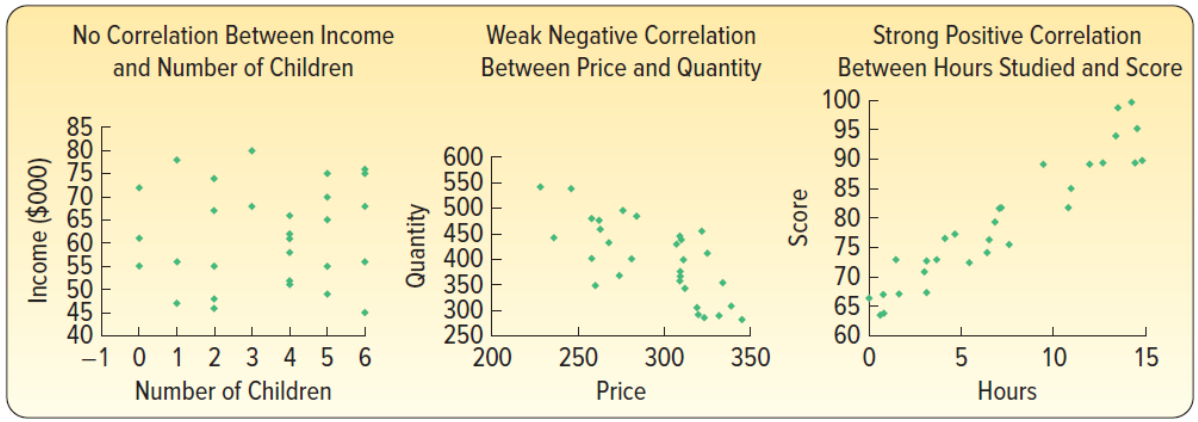
Correlation Coefficient (2 of 2)

- The following graphs summarize the strength and direction of the correlation coefficient





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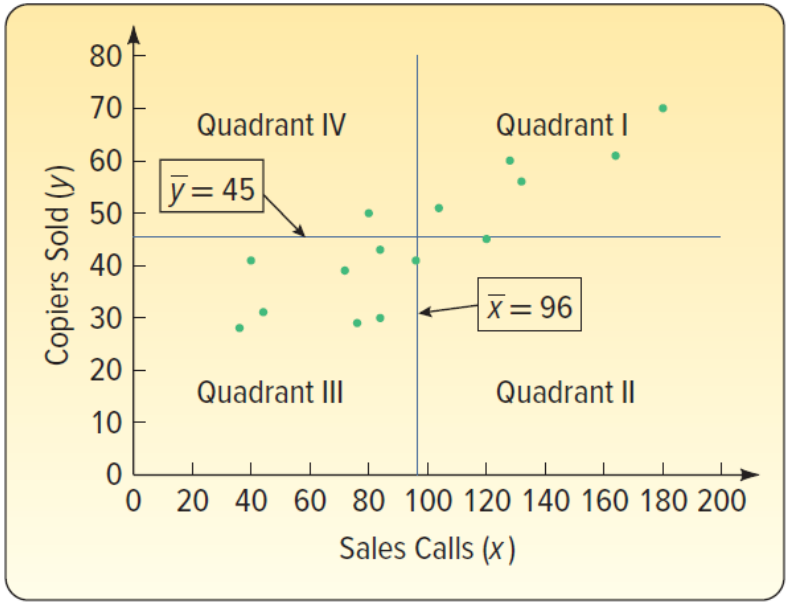
Correlation Coefficient, r

How is the correlation coefficient determined? We'll use North American Copier Sales as an example. We begin with a scatter diagram, but this time we'll draw a vertical line at the mean of the x-values (96 sales calls) and a horizontal line at the mean of the y-values (45 copiers).

Sales Representative	Sales Calls	Copiers Sold
Brian Virost	96	41
Carlos Ramirez	40	41
Carol Saia	104	51
Greg Fish	128	60
Jeff Hall	164	61
Mark Reynolds	76	29
Meryl Rumsey	72	39
Mike Kiel	80	50
Ray Snarsky	36	28
Rich Niles	84	43
Ron Broderick	180	70
Sal Spina	132	56
Soni Jones	120	45
Susan Welch	44	31
Tom Keller	84	30
Total	1440	675



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Correlation Coefficient, r, Continued

How is the correlation coefficient determined? Now we find the deviations from the mean number of sales calls and the mean number of copiers sold; then multiply them. The sum of their product is 6,672 and will be used in formula 13-1 to find r. We also need the standard deviations. The result, r=.865 indicates a strong, positive relationship.

CORRELATION COEFFICIENT

$$r = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{(n - 1)s_x s_y} \qquad [13\text{–}1]$$

Sales Representative	Sales Calls (x)	Copiers Sold (y)	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
Brian Virost	96	41	0	−4	0
Carlos Ramirez	40	41	−56	−4	224
Carol Saia	104	51	8	6	48
Greg Fish	128	60	32	15	480
Jeff Hall	164	61	68	16	1,088
Mark Reynolds	76	29	−20	−16	320
Meryl Rumsey	72	39	−24	−6	144
Mike Kiel	80	50	−16	5	−80
Ray Snarsky	36	28	−60	−17	1,020
Rich Niles	84	43	−12	−2	24
Ron Broderick	180	70	84	25	2,100
Sal Spina	132	56	36	11	396
Soni Jones	120	45	24	0	0
Susan Welch	44	31	−52	−14	728
Tom Keller	84	30	−12	−15	180
Totals	1440	675	0	0	6,672

	A	B	C	D	E	F	G	H
1		Sales Representative	Sales Calls (x)	Copiers Sold (y)			Sales Calls (x)	Copiers Sold (y)
2		Brian Virost	96	41		Mean	96.00	45.00
3		Carlos Ramirez	40	41		Standard Error	11.04	3.33
4		Carol Saia	104	51		Median	84.00	43.00
5		Greg Fish	128	60		Mode	84.00	41.00
6		Jeff Hall	164	61		Standard Deviation	42.76	12.89
7		Mark Reynolds	76	29		Sample Variance	1828.57	166.14
8		Meryl Rumsey	72	39		Kurtosis	-0.32	-0.73
9		Mike Kiel	80	50		Skewness	0.46	0.36
10		Ray Snarsky	36	28		Range	144.00	42.00
11		Rich Niles	84	43		Minimum	36.00	28.00
12		Ron Broderick	180	70		Maximum	180.00	70.00
13		Sal Spina	132	56		Sum	1440.00	675.00
14		Soni Jones	120	45		Count	15.00	15.00
15		Susan Welch	44	31				
16		Tom Keller	84	30				
17		Total	1440	675				

Source: Microsoft Excel

r = = 0.865

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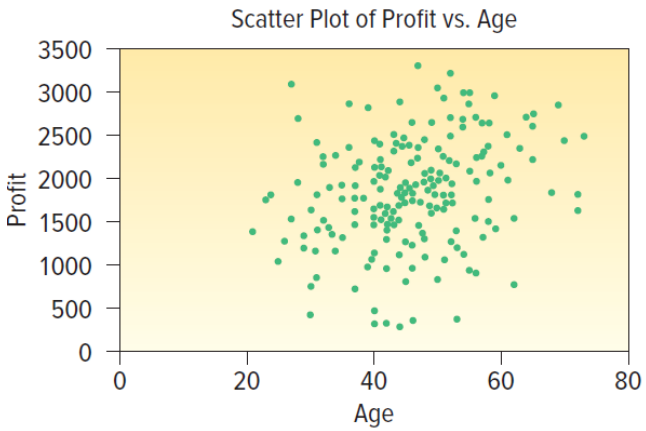
Correlation Coefficient Example

The Applewood Auto Group's marketing department believes younger buyers purchase vehicles on which lower profits are earned and older buyers purchase vehicles on which higher profits are earned. They would like to use this information as part of an upcoming advertising campaign to try to attract older buyers. Develop a scatter diagram and then determine the correlation coefficient. Would this be a useful advertising feature?



	A	B	C
1		***Age***	***Profit***
2	Age	1	
3	Profit	0.262	1

The scatter diagram suggests that a positive relationship does exist between age and profit, but it does not appear to be a strong relationship.

Next, calculate r, which is 0.262. The relationship is positive but weak. The data does not support a business decision to create an advertising campaign to attract older buyers!

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Testing the Significance of r

- Recall that the sales manager from North American Copier Sales found an r of 0.865
- Could the result be due to sampling error? Remember only 15 salespeople were sampled
- We ask the question, could there be zero correlation in the population from which the sample was selected?
- We'll let ρ represent the correlation in the population and conduct a hypothesis test to find out

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Testing the Significance of r Example

Step 1: State the null and the alternate hypothesis

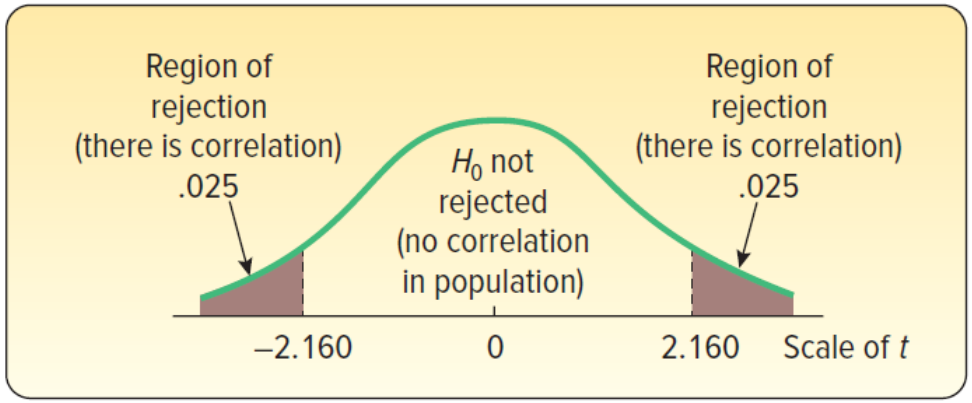
H_0: = 0 The correlation in the population is zero

H_1: ≠ 0 The correlation in the population is different from zero

Step 2: Select the level of significance; we'll use .05

Step 3: Select the test statistic; we use t

Step 4: Formulate the decision rule, reject H_0 if t < 2.160 or > 2.160



Decision Rule for Test of Hypothesis at .05 Significance Level and 13 *df*

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Testing the Significance of r Example Continued

Step 5: Make decision; reject H_0, t=6.216

t TEST FOR THE CORRELATION COEFFICIENT

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \text{ with } n-2 \text{ degrees of freedom} \qquad [13\text{–}2]$$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{.865\sqrt{15-2}}{\sqrt{1-.865^2}} = 6.216$$

Step 6: Interpret; there is correlation with respect to the number of sales calls made and the number of copiers sold in the population of salespeople.

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Testing the Significance of the Correlation Coefficient

In the Applewood Auto Group example, we found an r=0.262 which is positive, but rather weak. We test our conclusion by conducting a hypothesis test that the correlation is greater than 0.

Step 1: State the null and the alternate hypothesis

H_0: ≤ 0 The correlation in the population is negative or zero

H_1: > 0 The correlation in the population is positive

Step 2: Select the level of significance; we'll use .05

Step 3: Select the test statistic; we use t

Step 4: Formulate the decision rule, reject H_0 if $t > 1.653$

Step 5: Make decision; reject H_0, t=3.622

Step 6: Interpret; there is correlation with respect to profits and age of the buyer

We use formula (13–2) to find the value of the test statistic.

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.262\sqrt{180-2}}{\sqrt{1-0.262^2}} = 3.622$$

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Regression Analysis

- In regression analysis, we estimate one variable based on another variable
- The variable being estimated is the dependent variable
- The variable used to make the estimate or predict the value is the independent variable
- The relationship between the variables is linear
- Both the independent and the dependent variables must be interval or ratio scale

REGRESSION EQUATION An equation that expresses the linear relationship between two variables.

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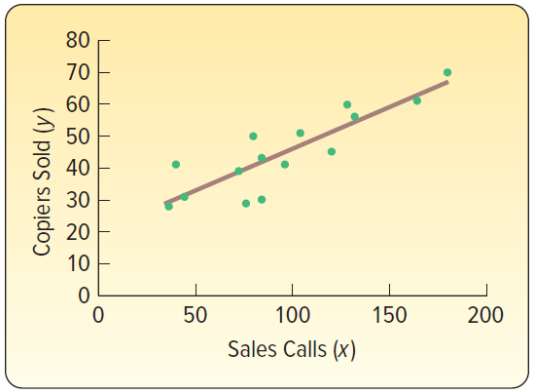
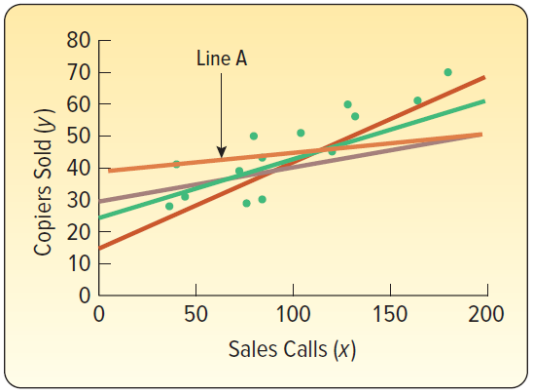
Least Squares Principle

- In regression analysis, our objective is to use the data to position a line that best represents the relationship between two variables
- The first approach is to use a scatter diagram to visually position the line





- But this depends on judgement; we would prefer a method that results in a single, best regression line

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Least Squares Regression Line

LEAST SQUARES PRINCIPLE A mathematical procedure that uses the data to position a line with the objective of minimizing the sum of the squares of the vertical distances between the actual y values and the predicted values of y.

- To illustrate, the same data are plotted in the three charts below

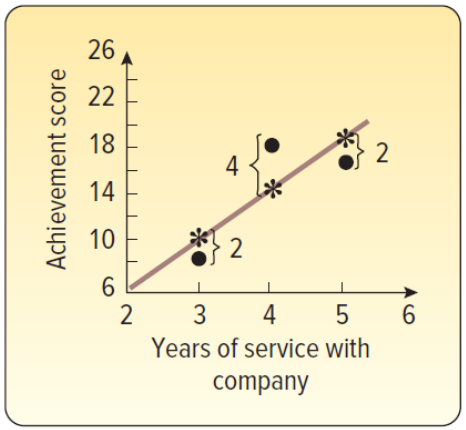

CHART 13–9 The Least Squares Line

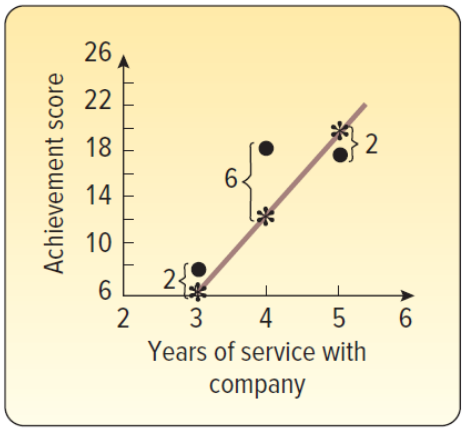

CHART 13–10 Line Drawn with a Straight Edge

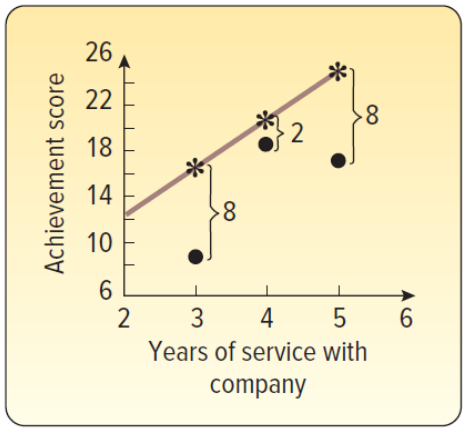

CHART 13–11 Different Line Drawn with a Straight Edge

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Least Squares Regression Line (2 of 2)

- This is the equation of a line

GENERAL FORM OF LINEAR REGRESSION EQUATION	$\hat{y} = a + bx$	**[13–3]**

  - $\hat{y}$ is the estimated value of y for a selected value of x
  - a is the constant or intercept
  - b is the slope of the fitted line
  - x is the value of the independent variable
- The formulas for a and b are

SLOPE OF THE REGRESSION LINE	$b = r\left(\frac{s_y}{s_x}\right)$	**[13–4]**

***Y*-INTERCEPT**	$a = \bar{y} - b\bar{x}$	**[13–5]**

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Least Squares Regression Line Example

Recall the example of North American Copier Sales. The sales manager gathered information on the number of sales calls made and the number of copiers sold. Use the least squares method to determine a linear equation to express the relationship between the two variables.

- The first step is to find the slope of the least squares regression line, b

$$b = r\left(\frac{s_y}{s_x}\right) = .865\left(\frac{12.89}{42.76}\right) = 0.2608$$

- Next, find a

$$a = \bar{y} - b\bar{x} = 45 - .2608(96) = 19.9632$$

- Then determine the regression line

$$\hat{y} = 19.9632 + 0.2608x$$

- So if a salesperson makes 100 calls, he or she can expect to sell 46.0432 copiers

$$\hat{y} = 19.9632 + 0.2608x = 19.9632 + 0.2608(100) = 46.0432$$

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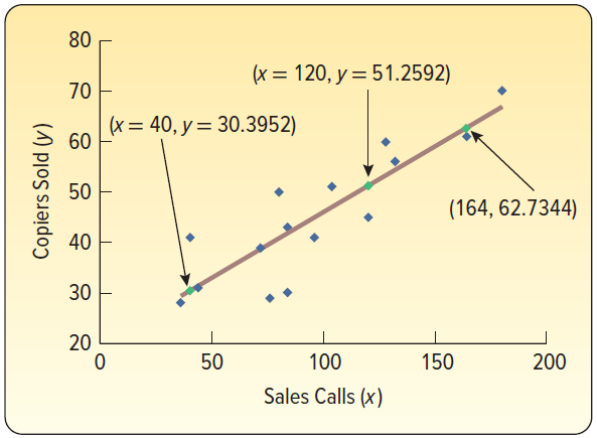
Drawing the Regression Line

The least squares equation can be drawn on the scatter diagram. For example, the fifth sales representative is Jeff Hall. He made 164 calls. His estimated number of copiers sold is 62.7344. The plot x = 164 and $\hat{y}$ = 62.7344 is located by moving to 164 on the x-axis and then going vertically to 63.7344. The other points on the regression equation can be determined by substituting a particular value of x into the regression equation and calculating $\hat{y}$.



$$\hat{y} = 19.9632 + 0.2608x$$

Sales Representative	Sales Calls (x)	Copiers Sold (y)	Estimated Sales ($\hat{y}$)
Brian Virost	96	41	45.0000
Carlos Ramirez	40	41	30.3952
Carol Saia	104	51	47.0864
Greg Fish	128	60	53.3456
Jeff Hall	164	61	62.7344
Mark Reynolds	76	29	39.7840
Meryl Rumsey	72	39	38.7408
Mike Kiel	80	50	40.8272
Ray Snarsky	36	28	29.3520
Rich Niles	84	43	41.8704
Ron Broderick	180	70	66.9072
Sal Spina	132	56	54.3888
Soni Jones	120	45	51.2592
Susan Welch	44	31	31.4384
Tom Keller	84	30	41.8704

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Regression Equation Slope Test

- For a regression equation, the slope is tested for significance
- We test the hypothesis that the slope of the line in the population is 0
- If we do not reject the null hypothesis, we conclude there is **no relationship** between the two variables
- When testing the null hypothesis about the slope, the test statistic is with n – 2 degrees of freedom
- We begin with the hypothesis statements

$$H_0: \ = 0$$

$$H_1: \ \neq 0$$

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Regression Equation Slope Test Example

Recall the North American Copier Sales example. We identified the slope as b and it is our estimate of the slope of the population, . We conduct a hypothesis test.

Step 1: State the null and alternate hypothesis

H_0: ≤ 0

H_1: > 0

Step 2: Select the level of significance; we use .05

Step 3: Select the test statistic, t

Step 4: Formulate the decision rule, reject H_0 if t > 1.771

Step 5: Make decision; reject H_0, t = 6.205

Step 6: Interpret; the number of sales calls is useful in estimating copier sales

TEST FOR THE SLOPE $t = \frac{b - 0}{s_b}$ with $n - 2$ degrees of freedom **[13–6]**

$$t = \frac{b - 0}{s_b} = \frac{0.2606 - 0}{0.042} = 6.205$$

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Regression Equation Slope Test Example (2 of 2)

	A	B	C	D	E	F	G	H	I	J
1	**Sales Representive**	**Sales calls (x)**	**Copiers Sold (y)**		SUMMARY OUTPUT					
2	Brian Virost	96	41							
3	Carlos Ramirez	40	41		*Regression Statistics*					
4	Carol Saia	104	51		Multiple R	0.865				
5	Greg Fish	128	60		R Square	0.748				
6	Jeff Hall	164	61		Adjusted R Square	0.728				
7	Mark Reynolds	76	29		Standard Error	6.720				
8	Meryl Rumsey	72	39		Observations	15				
9	Mike Kiel	80	50							
10	Ray Snarsky	36	28		ANOVA					
11	Rich Niles	84	43			*df*	*SS*	*MS*	*F*	*Significance F*
12	Ron Broderick	180	70		Regression	1	1738.89	1738.89	38.5031	3.19277E-05
13	Sal Spina	132	56		Residual	13	587.11	45.1623		
14	Sani Jones	120	45		Total	14	2326			
15	Susan Welch	44	31							
16	Tom Keller	84	30			*Coefficients*	*Standard Error*	*t Stat*	*P-value*	
17					Intercept	19.9800	4.389675533	4.55159	0.00054	
18					Sales calls (x)	0.2606	0.042001817	6.20509	3.2E-05	

Source: Microsoft Excel

Highlighted, b is .2606; the standard error is .0420

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Evaluating a Regression Equation's Ability to Predict

- Perfect prediction is practically impossible in almost all disciplines, including economics and business
- The North American Copier Sales example showed a significant relationship between sales calls and copier sales, the equation is

  Number of copiers sold = 19.9632 + .2608(Number of sales calls)

- What if the number of sales calls is 84, and we calculate the number of copiers sold is 41.8704—we did have two employees with 84 sales calls, they sold just 30 and 24
- So, is the regression equation a good predictor?
- We need a measure that will tell how inaccurate the estimate might be

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The Standard Error of Estimate

- The standard error of estimate measures the variation around the regression line

STANDARD ERROR OF ESTIMATE A measure of the dispersion, or scatter, of the observed values around the line of regression for a given value of x.

- It is in the same units as the dependent variable
- It is based on squared deviations from the regression line
- Small values indicate that the points cluster closely about the regression line
- It is computed using the following formula

STANDARD ERROR OF ESTIMATE

$$s_{y \cdot x} = \sqrt{\frac{\Sigma(y - \hat{y})^2}{n - 2}} \qquad [13\text{–}7]$$

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The Standard Error of Estimate Example

We calculate the standard error of estimate in this example. We need the sum of the squared differences between each observed value of y and the predicted value of y, which is . We use a spreadsheet to help with the calculations.

	A	B	C	D	E	F
1	**Sales Rep**	**Sales Calls (x)**	**Copiers Sold (y)**	**Estimated Sales**	**$(y - \hat{y})$**	**$(y - \hat{y})^2$**
2	Brian Virost	96	41	45.0000	–4.0000	16.0000
3	Carlos Ramirez	40	41	30.3952	10.6048	112.4618
4	Carol Saia	104	51	47.0864	3.9136	15.3163
5	Greg Fish	128	60	53.3456	6.6544	44.2810
6	Jeff Hall	164	61	62.7344	–1.7344	3.0081
7	Mark Reynolds	76	29	39.7840	–10.7840	116.2947
8	Meryl Rumsey	72	39	38.7408	0.2592	0.0672
9	Mike Kiel	80	50	40.8272	9.1728	84.1403
10	Ray Snarsky	36	28	29.3520	–1.3520	1.8279
11	Rich Niles	84	43	41.8704	1.1296	1.2760
12	Ron Broderick	180	70	66.9072	3.0928	9.5654
13	Sal Spina	132	56	54.3888	1.6112	2.5960
14	Soni Jones	120	45	51.2592	–6.2592	39.1776
15	Susan Welch	44	31	31.4384	–0.4384	0.1922
16	Tom Keller	84	30	41.8704	–11.8704	140.9064
17	**Total**				**0.0000**	**587.1108**

The standard error of estimate is 6.720

$$s_{y \cdot x} = \sqrt{\frac{\Sigma(y - \hat{y})^2}{n - 2}} = \sqrt{\frac{587.1108}{15 - 2}} = 6.720$$

If the standard error of estimate is small, this indicates that the data are relatively close to the regression line and the regression equation can be used. If it is large, the data are widely scattered around the regression line and the regression equation will not provide a precise estimate of y.

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Coefficient of Determination

COEFFICIENT OF DETERMINATION The proportion of the total variation in the dependent variable Y that is explained, or accounted for, by the variation in the independent variable X.

- It ranges from 0 to 1.0
- It is the square of the correlation coefficient
- It is found from the following formula

COEFFICIENT OF DETERMINATION

$$r^2 = \frac{\text{SSR}}{\text{SS Total}} = 1 - \frac{\text{SSE}}{\text{SS Total}} \qquad [13\text{–}8]$$

- In the North American Copier Sales example, the correlation coefficient was .865; just square that $(.865)^2 = .748$; this is the coefficient of determination
- This means 74.8% of the variation in the number of copiers sold is explained by the variation in sales calls

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Relationships among r, r_2, and $s_{y,x}$

- Recall the **standard error of estimate** measures how close the actual values are to the regression line
  - When it is small, the two variables are closely related
- The **correlation coefficient** measures the strength of the linear association between two variables
  - When points on the scatter diagram are close to the line, the correlation coefficient tends to be large
- Therefore, the correlation coefficient and the standard error of estimate are inversely related

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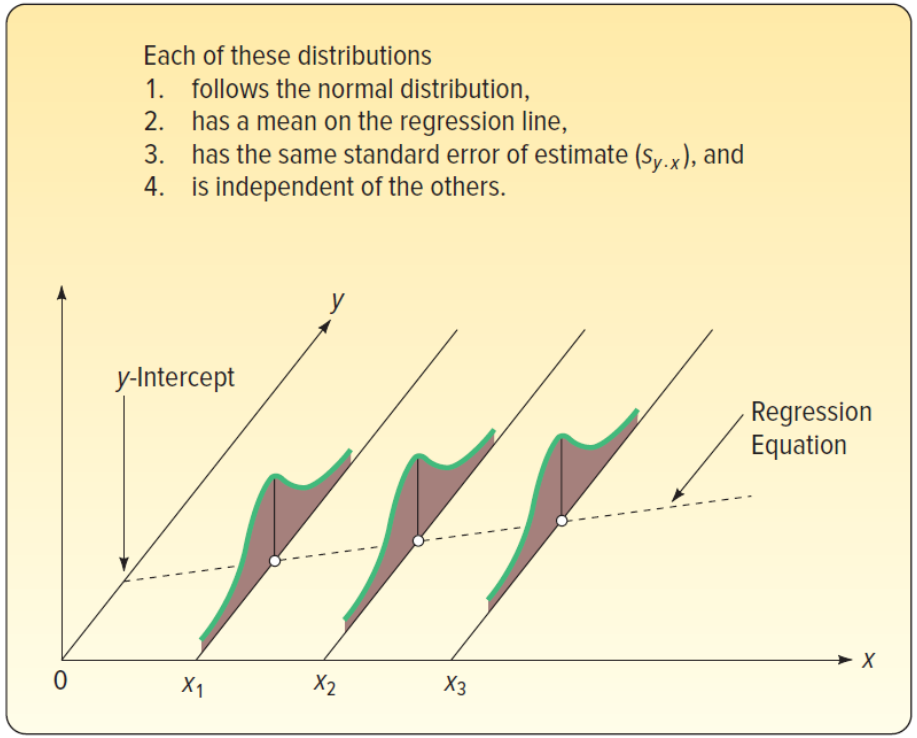
Inference about Linear Regression

- We can predict the number of copiers sold (y) for a selected value of number of sales calls made (x)
- But first, let's review the regression assumptions of each of the distributions in the graph below



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Constructing Confidence and Prediction Intervals

- Use a confidence interval when the regression equation is used to predict the mean value of y for a given value of x
- For instance, we would use a confidence interval to estimate the mean salary of all executives in the retail industry based on their years of experience

CONFIDENCE INTERVAL FOR THE MEAN OF Y, GIVEN X

$$\hat{y} \pm t s_{y \cdot x} \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{\Sigma (x - \bar{x})^2}} \qquad [13\text{–}11]$$

- Use a prediction interval when the regression equation is used to predict an individual y for a given value of x
- For instance, we would estimate the salary of a particular retail executive who has 20 years of experience

PREDICTION INTERVAL FOR Y, GIVEN X

$$\hat{y} \pm t s_{y \cdot x} \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{\Sigma (x - \bar{x})^2}} \qquad [13\text{–}12]$$

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Confidence Interval and Prediction Interval Example

We return to the North American Copier Sales example. Determine a 95% confidence interval for all sales representatives who make 50 calls, and determine a prediction interval for Sheila Baker, a west coast sales representative who made 50 sales calls.

Sales Representative	Sales Calls (x)	Copiers Sold (y)	$(x - \bar{x})$	$(x - \bar{x})^2$
Brian Virost	96	41	0	0
Carlos Ramirez	40	41	−56	3,136
Carol Saia	104	51	8	64
Greg Fish	128	60	32	1,024
Jeff Hall	164	61	68	4,624
Mark Reynolds	76	29	−20	400
Meryl Rumsey	72	39	−24	576
Mike Kiel	80	50	−16	256
Ray Snarsky	36	28	−60	3,600
Rich Niles	84	43	−12	144
Ron Broderick	180	70	84	7,056
Sal Spina	132	56	36	1,296
Sani Jones	120	45	24	576
Susan Welch	44	31	−52	2,704
Tom Keller	84	30	−12	144
Total	1440	675	0	25,600

$$\hat{y} = 19.9632 + 0.2608x = 19.9632 + 0.2608\,(50) = 33.0032$$

$$\text{Confidence Interval} = \hat{y} \pm t s_{y \cdot x}\sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{\Sigma(x - \bar{x})^2}}$$

$$= 33.0032 \pm 2.160(6.720)\sqrt{\frac{1}{15} + \frac{(50 - 96)^2}{25{,}600}}$$

$$= 33.0032 \pm 5.6090$$

$$\text{Prediction Interval} = \hat{y} \pm t s_{y \cdot x}\sqrt{1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{\Sigma(x - \bar{x})^2}}$$

$$= 33.0032 \pm 2.160(6.720)\sqrt{1 + \frac{1}{15} + \frac{(50 - 96)^2}{25{,}600}}$$

$$= 33.0032 \pm 15.5612$$

The 95% confidence interval for all sales representatives is 27.3942 up to 38.6122. The 95% prediction interval for Sheila Baker is 17.442 up to 48.5644 copiers.

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Transforming Data

- Regression analysis and the correlation coefficient requires data to be linear
- But what if data is not linear?
- If data is not linear, we can rescale one or both of the variables so the new relationship is linear
- Common transformations include
  - Computing the log to the base 10 of y, Log(y)
  - Taking the square root
  - Taking the reciprocal
  - Squaring one or both variables
- Caution: when you are interpreting a correlation coefficient or regression equation – it could be nonlinear

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Transforming Data Example

GroceryLand Supermarkets is a regional grocery chain located in the midwestern United States. The director of marketing wishes to study the effect of price on weekly sales of their two-liter private brand diet cola. The objectives of the study are

1. To determine whether there is a relationship between selling price and weekly sales. Is this relationship direct or indirect? Is it strong or weak?
2. To determine the effect of price increases or decreases on sales. Can we effectively forecast sales based on the price?

To begin, the company decides to price the two-liter diet cola from $0.50 to $2.00. To collect the data, a random sample of 20 stores is taken and then each store is randomly assigned a selling price.

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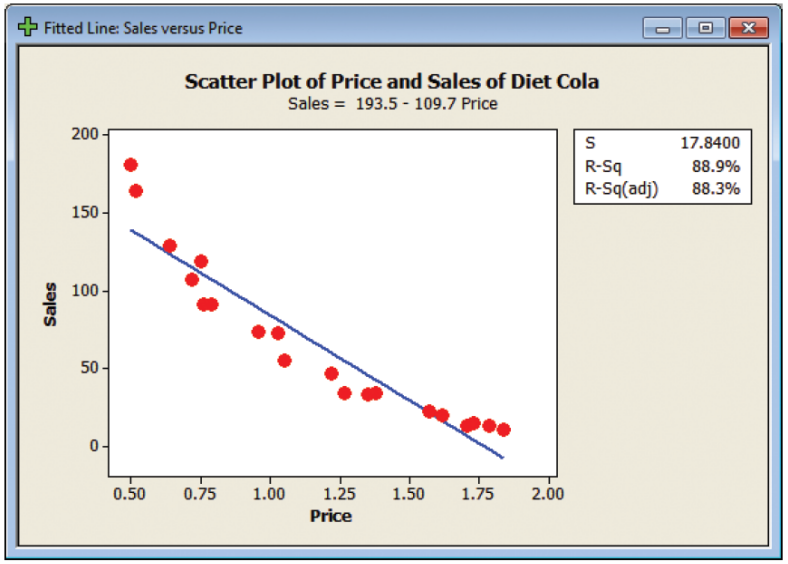
Transforming Data Example (2 of 3)

GroceryLand Sales and Price Data		
Store Number	Price	Sales
A-17	0.50	181
A-121	1.35	33
A-227	0.79	91
A-135	1.71	13
A-6	1.38	34
A-282	1.22	47
A-172	1.03	73
A-296	1.84	11
A-143	1.73	15
A-66	1.62	20

GroceryLand Sales and Price Data		
Store Number	Price	Sales
A-30	0.76	91
A-127	1.79	13
A-266	1.57	22
A-117	1.27	34
A-132	0.96	74
A-120	0.52	164
A-272	0.64	129
A-120	1.05	55
A-194	0.72	107
A-105	0.75	119



A strong, inverse relationship!

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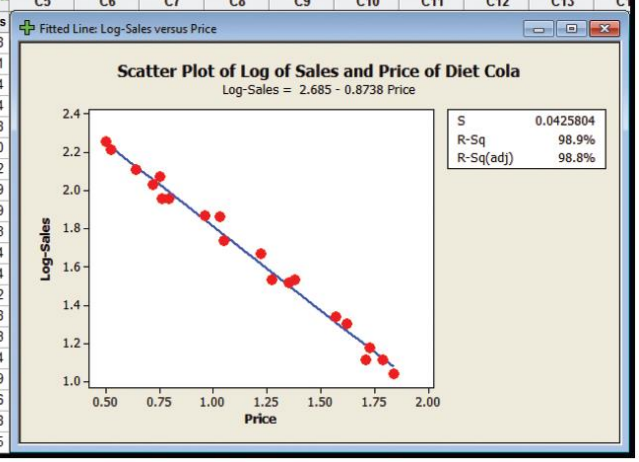
Transforming Data Example (3 of 3)

The director of marketing decides to transform the dependent variable, Sales, by taking the logarithm to the base 10 of each sales value. Note the new variable, Log-Sales, in the following analysis as it is used as the dependent variable with Price as the independent variable.

Groceryland.MTW ***

	C1	C2	C3	C4
	Store Number	Price	Sales	Log-Sales
1	17	0.50	181	2.25768
2	121	1.35	33	1.51851
3	227	0.79	91	1.95904
4	135	1.71	13	1.11394
5	6	1.38	34	1.53148
6	282	1.22	47	1.67210
7	172	1.03	73	1.86332
8	296	1.84	11	1.04139
9	143	1.73	15	1.17609
10	66	1.62	20	1.30103
11	30	0.76	91	1.95904
12	127	1.79	13	1.11394
13	266	1.57	22	1.34242
14	117	1.27	34	1.53148
15	132	0.96	74	1.86923
16	120	0.52	164	2.21484
17	272	0.64	129	2.11059
18	120	1.05	55	1.74036
19	194	0.72	107	2.02938
20	105	0.75	119	2.07555



1. By transforming the dependent variable, Sales, we increase the coefficient of determination from 0.889 to 0.989. So now Price explains almost all of the variation in Log-Sales.
2. The transformed data "fit" the linear relationship better.
3. The regression equation is $= 2.685 - .8738x$.

$= 2.685 - .8738(x) = 2.685 - .8738\ (1.25) = 1.593$

Now, undo the transformation by taking the antilog of 1.593; 39.174 or 39 bottles. That is, if they price the cola at \$1.25, they'll sell 39 bottles; at \$2.00 they'll sell 9.

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Chapter 12 Practice Problems

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Question 3

LO13-2

Bi-lo Appliance Super-Store has outlets in several large metropolitan areas in New England. The general sales manager aired a commercial for a digital camera on selected local TV stations prior to a sale starting on Saturday and ending Sunday. She obtained the information for Saturday–Sunday digital camera sales at the various outlets and paired it with the number of times the advertisement was shown on the local TV stations. The purpose is to find whether there is any relationship between the number of times the advertisement was aired and digital camera sales. The pairings are:

a. What is the dependent variable?
b. Draw a scatter diagram.
c. Determine the correlation coefficient.
d. Interpret these statistical measures.

Location of TV Station	Number of Airings	Saturday–Sunday Sales ($ thousands)
Providence	4	15
Springfield	2	8
New Haven	5	21
Boston	6	24
Hartford	3	17

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Question 11

LO13-2

The Airline Passenger Association studied the relationship between the number of passengers on a particular flight and the cost of the flight. It seems logical that more passengers on the flight will result in more weight and more luggage, which in turn will result in higher fuel costs. For a sample of 15 flights, the correlation between the number of passengers and total fuel cost was .667. Is it reasonable to conclude that there is positive association in the population between the two variables? Use the .01 significance level.

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Question 17

LO13-3

Bloomberg Intelligence listed 50 companies to watch in 2018 (www.bloomberg.com/features/companies-to-watch-2018). Twelve of the companies are listed here with their total assets and 12-month sales.

Company	Total Assests ($ Billions)	12-Month Sales ($ Billions)	Company	Total Assests ($ Billions)	12-Month Sales ($ Billions)
Eni SpA	134.47	68.13	AMC Entertainment	9.81	4.19
Discovery Communications	16.15	6.59	DSV	6.14	10.54
Dong Energy	20.5	9.16	Alaska Air	10.7	6.9
AMD	3.37	4.62	Boohoo.com	0.235	0.545
Cabot Oil & Gas	5.22	1.58	Danone	52.12	25.11
Engie	166.84	72	Experian	7.69	4.34

Let sales be the dependent variable and total assets the independent variable.

a. Draw a scatter diagram.

b. Compute the correlation coefficient.

c. Determine the regression equation.

d. For a company with $100 billion in assets, predict the 12-month sales.

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Question 23

LO13-4

Refer to Exercise 17. The regression equation is $\hat{y} = 1.85 + .08x$, the sample size is 12, and the standard error of the slope is 0.03. Use the .05 significance level. Can we conclude that the slope of the regression line is *different from zero*?

Company	Total Assests ($ Billions)	12-Month Sales ($ Billions)	Company	Total Assests ($ Billions)	12-Month Sales ($ Billions)
Eni SpA	134.47	68.13	AMC Entertainment	9.81	4.19
Discovery Communications	16.15	6.59	DSV	6.14	10.54
Dong Energy	20.5	9.16	Alaska Air	10.7	6.9
AMD	3.37	4.62	Boohoo.com	0.235	0.545
Cabot Oil & Gas	5.22	1.58	Danone	52.12	25.11
Engie	166.84	72	Experian	7.69	4.34

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Question 27

LO13-5

Bradford Electric Illuminating Company is studying the relationship between kilowatt-hours (thousands) used and the number of rooms in a private single-family residence. A random sample of 10 homes yielded the following:

Number of Rooms	Kilowatt-Hours (thousands)	Number of Rooms	Kilowatt-Hours (thousands)
12	9	8	6
9	7	10	8
14	10	10	10
6	5	5	4
10	8	7	7

Determine the standard error of estimate and the coefficient of determination. Interpret the coefficient of determination.

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Question 33

LO13-6

Bradford Electric Illuminating Company is studying the relationship between kilowatt-hours (thousands) used and the number of rooms in a private single-family residence. A random sample of 10 homes yielded the following:

Number of Rooms	Kilowatt-Hours (thousands)	Number of Rooms	Kilowatt-Hours (thousands)
12	9	8	6
9	7	10	8
14	10	10	10
6	5	5	4
10	8	7	7

a. Determine the .95 confidence interval, in thousands of kilowatt-hours, for the mean of all six-room homes.

b. Determine the .95 prediction interval, in thousands of kilowatt-hours, for a particular six-room home.

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Question 35

LO13-7

Using the following data with *x* as the independent variable and *y* as the dependent variable, answer the items.

x	−8	−16	12	2	18
y	58	247	153	3	341

a. Create a scatter diagram and describe the relationship between *x* and *y*.
b. Compute the correlation coefficient.
c. Transform the *x* variable by squaring each value, *x*2.
d. Create a scatter diagram and describe the relationship between *x*2 and *y*.
e. Compute the correlation coefficient between *x*2 and *y*.
f. Compare the relationships between *x* and *y*, and *x*2 and *y*.
g. Interpret your results.

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